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PERFORMANCE SPECIFICATION

GREASE, FOOD GRADE, FOOD PROCESSING EQUIPMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers greases for use in food processing equipment in applications where there may be incidental contact with food.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this listed, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise-specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9150

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D97 - Standard Test Method for Pour Point of Petroleum Products

ASTM D217 - Standard Test Method for Cone Penetration of Lubricating Grease

ASTM D1743 - Standard Test Method for Determining Corrosion Preventive Properties of Lubricating Greases

ASTM D2265 - Standard Test Method for Dropping Point of Lubricating Grease Over Wide Temperature Range

(Copies of these documents are available online at www.astm.org.)

NATIONAL SANITATION FOUNDATION (NSF)

Nonfood Compounds: List of Product Category Codes

(Copies of this document are available online at https://www.nsf.org.)

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Material Target List

(Copies of this document are available online at www.aia-aerospace.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The grease shall be a stable homogeneous blend of virgin petroleum oil and thickening agent plus additives, if necessary, to meet the requirements of this specification.

3.2 Chemical and physical properties. The grease shall have the chemical and physical properties specified in table I.

TABLE I. Chemical and physical requirements.

Property	Requirement
Penetration, worked	310-340
Base oil pour point, °C, max	-12
Dropping point, °C, min	149
Corrosion protection	Pass
NSF rating	H-1

3.3 National sanitation foundation (NSF) rating. Grease shall be listed in NSF under nonfood compounds and rated H-1 lubricants approved for incidental contact with food.

3.4 Prohibited materials. The grease shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

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3.5 Toxicity. When evaluated in accordance with 4.7, the grease shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.7 and 6.3).

3.6 Odor. The grease shall not have an objectionable odor or odor of rancidity or perfume.

4. VERIFICATION

4.1 Conformance inspection. Grease shall be inspected and tested to verify compliance with the requirements specified herein. Conformance inspection shall include the examination of 4.4 and the tests of 4.5 through 4.8.

4.2 Lot. For purposes of sampling, a lot shall consist of grease manufactured as one batch.

4.3 Sample material for testing. From each lot, three separate 2.27-kilogram sample specimens shall be selected, sealed, and marked. Two of the 2.27-kilogram samples shall be selected and forwarded to the testing laboratory acceptable to the Command or agency concerned. The final 2.27-kilogram sample shall be retained by the manufacturer or contractor for additional testing, if required, at a later date.

4.4 Examination of filled containers. Sample of filled containers selected from each lot in accordance with MIL-STD-1916 as specified (see 6.2) shall be examined to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests. Containers shall be examined for defects of the container and the closure, for evidence of leakage, and for unacceptable markings. Each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects or containing less product than stated on the container (under required fill) shall be cause for rejection of the container. If the number of defective containers in any sample exceeds the acceptable number for the appropriate sampling plan specified (see 6.2), this shall be cause for rejection of the lot represented by the sample.

4.4.1 Examination of compound material. Portions of the compound material shall be taken from the top, bottom, and intermediate parts of the sample of containers and examined visually to determine that there is no apparent difference in appearance or consistency. The material shall be homogeneous and show a smooth greasy consistency without lumps, crusts, or granular/crystalline particles. It shall not have an odor of rancidity or perfume.

4.4.2 Acceptance criteria. From each lot, containers shall be selected sealed, marked and forwarded to a testing laboratory acceptable to the Command or agency concerned. The selected containers shall meet the following criteria:

a. The sample specimens shall be subjected separately to the tests specified in 4.5.

b. If either specimen fails one or more of the tests specified in 4.5, the lot shall be rejected.

4.5. Testing methods. Test methods for requirements shall be in accordance with table II.

TABLE II. Chemical and physical test methods.

Property	Test Method or Verification
Penetration, worked	ASTM D217
Base oil pour point, °C, max	ASTM D97
Dropping point, °C, min	ASTM D2265
Corrosion protection	ASTM D1743
NSF rating	NSF Food Grade Authorization Program, Nonfood Compounds Category

4.6 Prohibited materials. Prohibited materials shall be verified as specified (see 3.4 and 6.2).

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4.7 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.5 as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the grease using data provided by the manufacturer/distributor to the NMCPHC (see 3.5 and 6.3).

4.8 Odor. The odor of the grease shall be observed immediately upon opening the test container (see 3.6).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Grease covered by this specification is intended for lubricant in food processing equipment. Current stock of grease may be used until depleted; future requisitions for grease should refer to the current version of this specification. Restrictions on composition, toxicity, and hazardous materials covered by this specification cause the grease to be military unique.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Requirement for prohibited materials verification (see 3.4 and 4.6).

c. Requirements for an HHA (see 3.5 and 4.7).

d. Sampling plan for the examination of filled containers and acceptable number of defective containers (see 4.4).

e. Packaging requirements (see 5.1).

f. When a Safety Data Sheet is required (see 6.4).

6.3 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, https://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.4 Safety data sheet. When specified (see 6.2), the contracting activity should be provided a Safety Data Sheet (SDS) at the time of contract award. The SDS is form OSHA-20 and found as part of FED-STD-313. The SDS should be included with each shipment of the material covered by this specification.

6.5 Subject term (key word) listing.

Nonfood compounds

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH
Air Force – 68

Review activity
DLA – GS

Preparing activity:
Navy – SH
(Project 9150-2020-010)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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